Decimal to Binary Conversion

$r = 2$

» To convert decimal to any other base 'r', divide integer part by r and multiply fractional part by r.

1) $(13)_{10} \longrightarrow (1101)_2$

$13 = 8 + 4 + 1$

1	1	0	1
b_3	b_2	b_1	b_0
↑	↑	↑	↑
2^3	2^2	2^1	2^0

$2^0 = 1$

$2^1 = 2$

$2^2 = 4$

$2^3 = 8$

$\times\ 2^4 = 16 > 13$

2) $(13)_{10} \longrightarrow (1101)_2$

Divisor ↓   Dividend ↓   rem

Divisor	Dividend	rem
2	13	1 → lsb
2	6	0
2	3	1
2	1	1 ← msb
2	0	0
	0	0

1 0 1 1 ~~X~~

1 1 0 1

$$\overset{I}{25}\,.\,\overset{F}{625}$$

$$(25.625)_{10} \longrightarrow (\qquad)_2$$

H.W.

i) $(67)_{10} \longrightarrow (\qquad)_2$

ii) $(129.75)_{10} \longrightarrow (\qquad)_2$

2	25	1 (lsb)
2	12	0
2	6	0
2	3	1
2	1	1 (msb)
	0	

$$00(11001.101)00_2$$

$$0.625 \times 2 = 1.25$$
$$0.25 \times 2 = 0.50$$
$$0.50 \times 2 = 1.00$$
$$0.00 \times 2 = 0.00$$

1) $(112)_{10} \longrightarrow (160)_8$

2) $(25.625)_{16} \longrightarrow (31.50)_8$

lsb

8	112	0
8	14	6
8	1	1
	0	

msb

160

lsb

8	25	1
8	3	3
	0	

msb

31.50

*Decimal to Octal

$0.625 \times 8 = 5.000$

$0.000 \times 8 = 0.000$

⋮

H.W.

i) $(2048)_{10} \longrightarrow (\quad)_8$

ii) $(82.250)_{10} \longrightarrow (\quad)_8$

$0.000 \times 8 = 0.000$

$\vdots$

Decimal to Hexadecimal Conversion

$r = 16$

>> For decimal to hexadecimal conversion, divide integer part by 16 and multiply fractional part by 16.

1) $(254)_{10} \longrightarrow (FE)_{16}$

$16 \times 16 = 240$

16	254	~~14~~ E
16	15	~~15~~ F
	0	

lsb

msb

2) $(25.625)_{10} \longrightarrow (19.A)_{16}$

16	25	9
16	1	1
	0	

$0.625 \times 16 = \cancel{10}.000$ A

$0.000 \times 16 = 0.000$

0
⋮
9

A	10
B	11
C	12
D	13
E	14
F	15

H.W. $(25.625)_{10} \longrightarrow (\quad)_4$

3) $(27.4)_{10} \longrightarrow (123.1212)_4 \quad r = 4$

4	27	3
4	6	2
4	1	1
	0	

lsb

msb

$0.4 \times 4 = 1.6$

$0.6 \times 4 = 2.4$

$0.4 \times 4 = 1.6$

$0.6 \times 4 = 2.4$

1212 12 ...